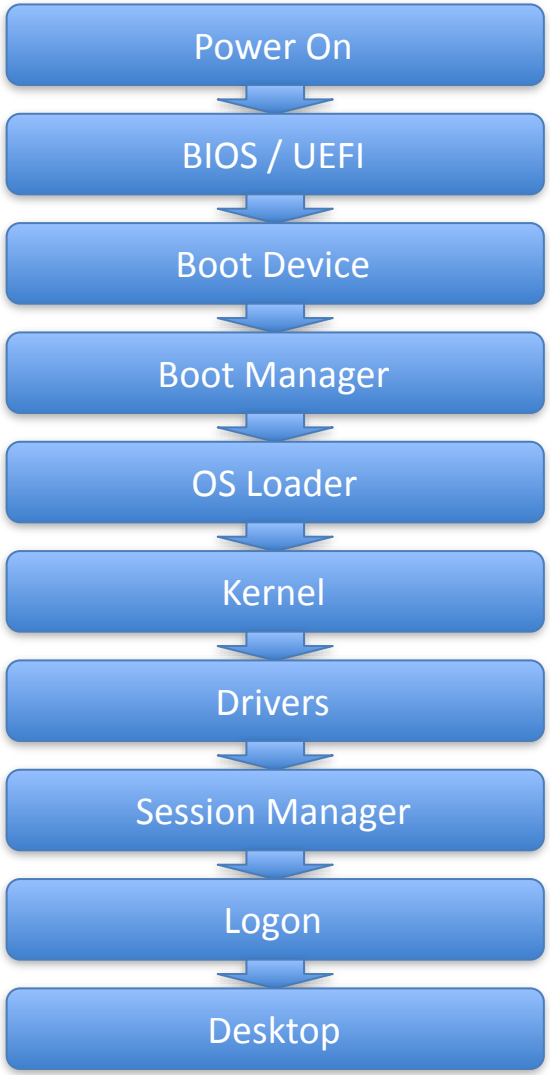
Windows Boot Sequence



1. BIOS / UEFI

- Performs the Power-On Self-Test (POST).
- Initializes CPU, memory, keyboard, storage and other hardware.
- Searches for a bootable storage device.
- Transfers control to the boot loader.

2. Windows Boot Manager & OS Loader

- Boot Manager reads the Boot Configuration Data (BCD).
- Allows OS selection if multiple operating systems are installed.
- winload.efi loads the Windows kernel.
- Loads the Hardware Abstraction Layer (HAL) and boot drivers.

3. Kernel Initialization & Logon

- The Windows kernel initializes memory and process management.
- Device drivers and core services are started.
- Session Manager starts essential Windows processes.
- winlogon.exe displays the login screen.
- explorer.exe loads the Windows desktop.

Summary

- Firmware initializes hardware.
- Boot Manager locates Windows.
- OS Loader loads the kernel and drivers.
- Kernel starts services and the login process.
- Desktop appears and Windows is ready to use.